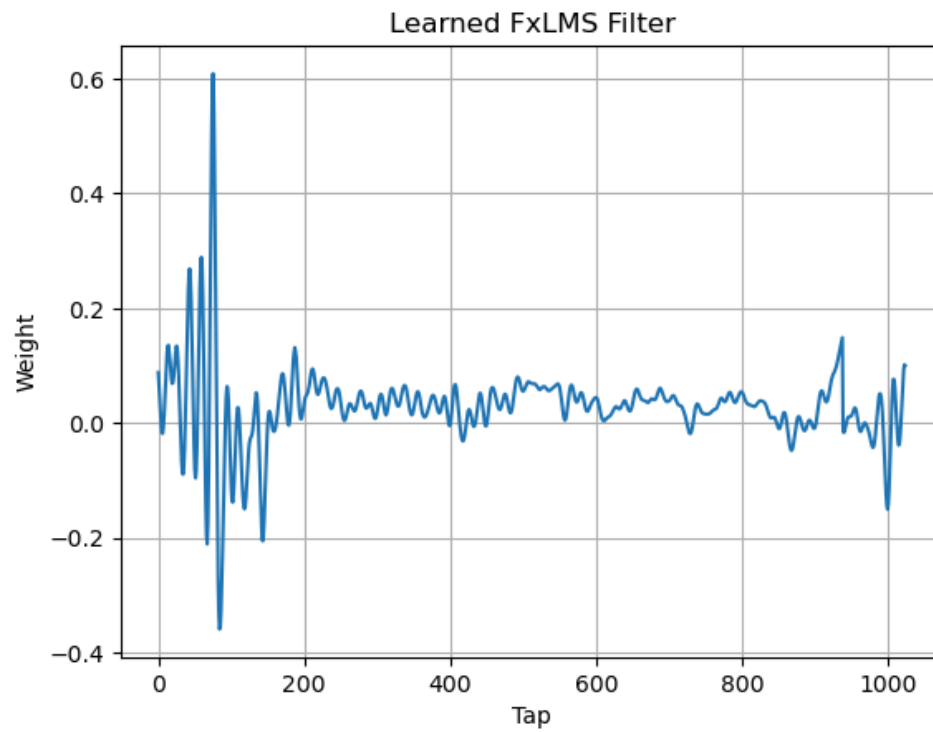
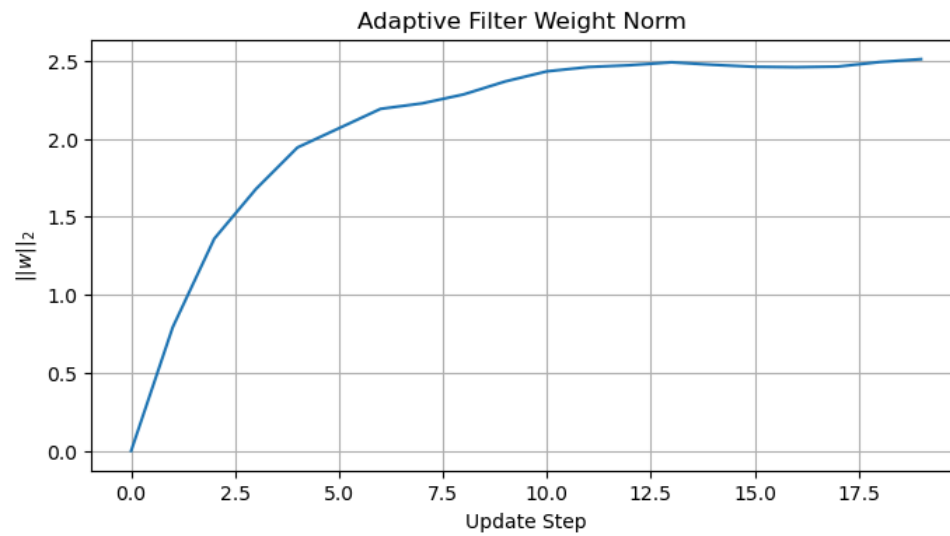


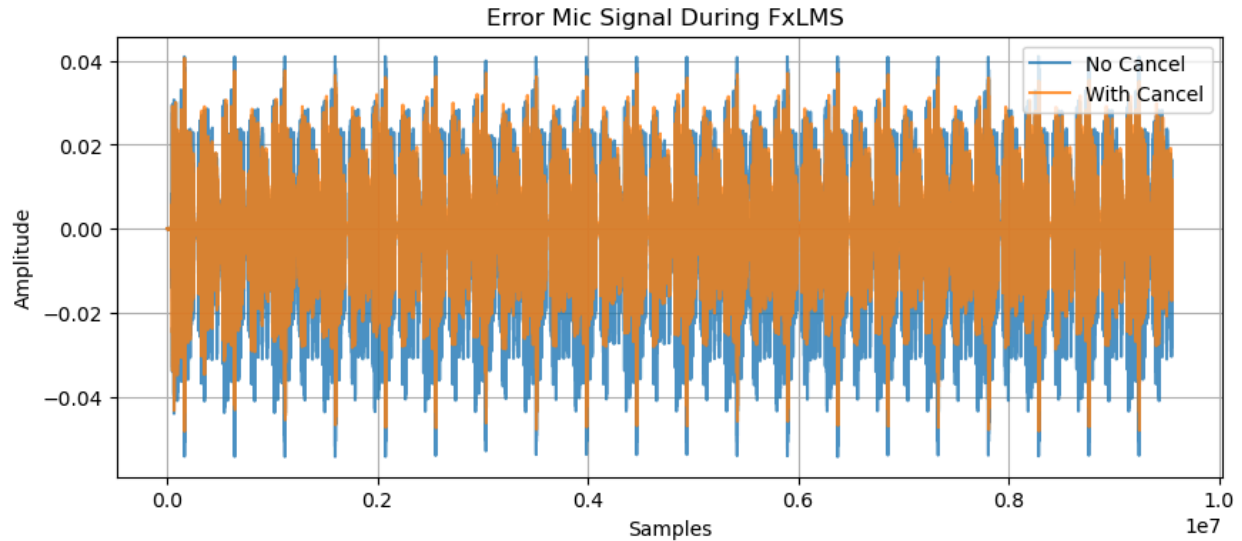
Weight logging and seeding

- Pulls weights in the background so playback loop isn't interrupted
- Send the RMS $\in [0, 1]$ then convert to norm in Python



Longer tests

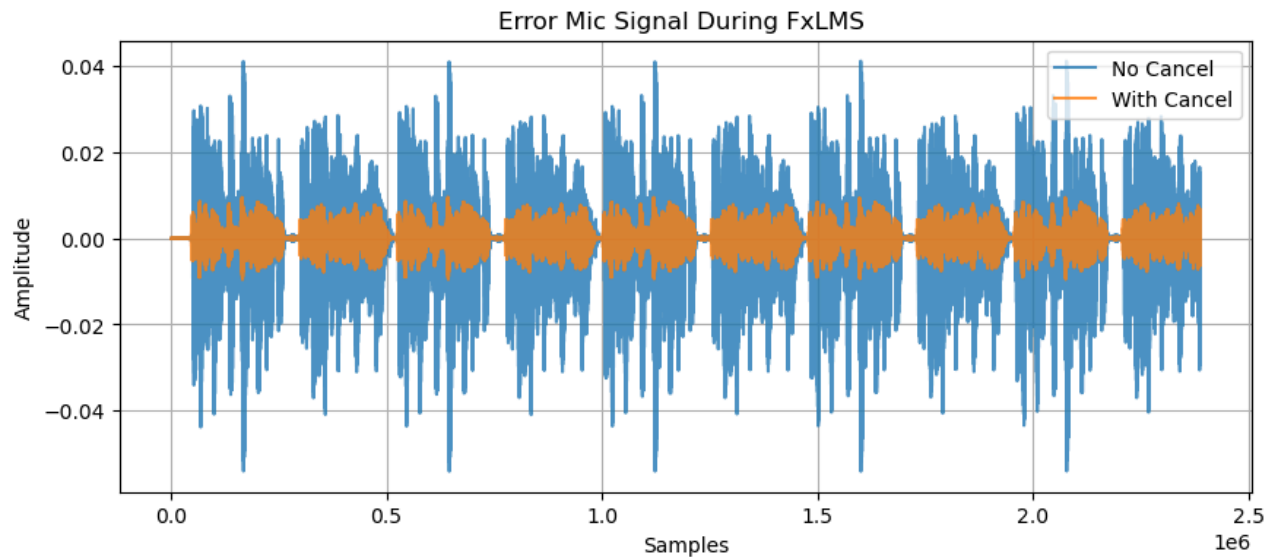
- -3.35dB



- Why does it cancel negative amplitudes better than positive?
- Why only -3dB?

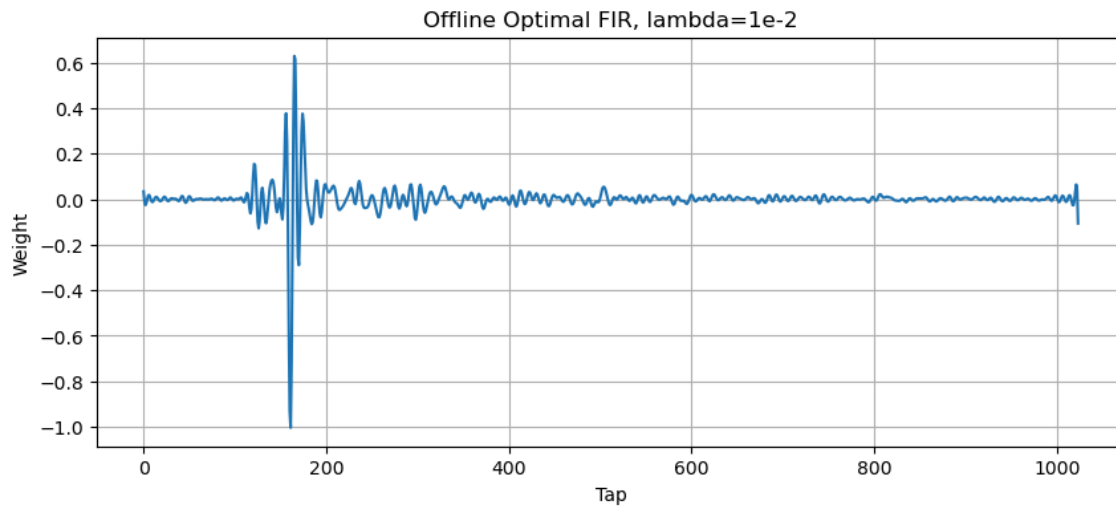
No Panel

- No stand: -9.92dB
- Stand but no panel: -9.78dB



Optimal offline filter

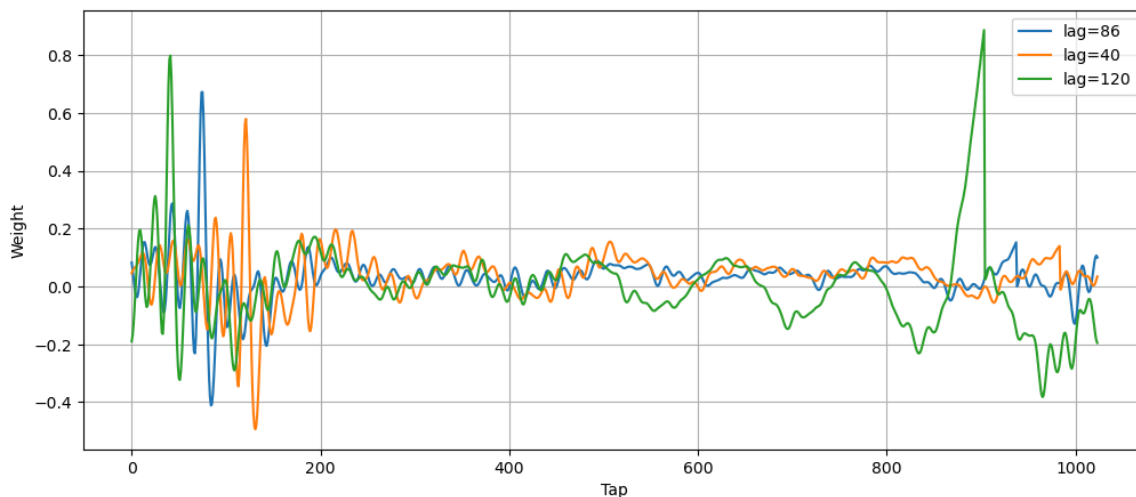
- Found the least squares optimal filter to transform ref mic to error mic
 - $r = e[n] + X_f w$,
 - $e[n]$ is the error mic recording, X_f if the ref mic filtered by the panel IR
 - $\min_w ||e + X_f w||^2$
- -18dB (we can cancel basically all reflections)
- w_{norm} was massive, so added L2 regularization
 - $\min_w ||e + X_f w||^2 + \lambda ||w||^2$
- Still -18dB



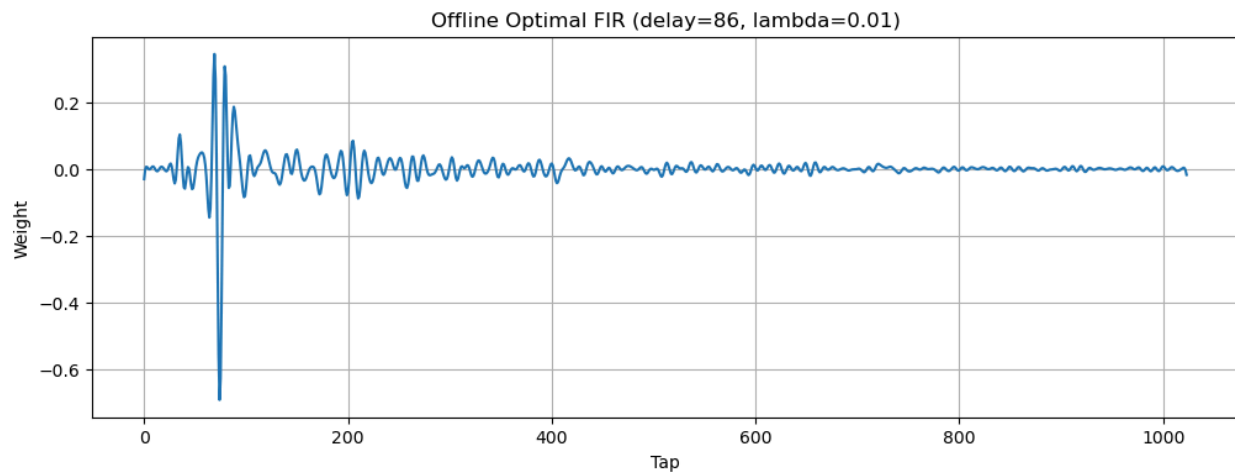
- When seeded with this filter, we only got like -.6dB

Lag sweep

- Does the lag affect where the peak of the filter lands?
- 0 lag was unstable

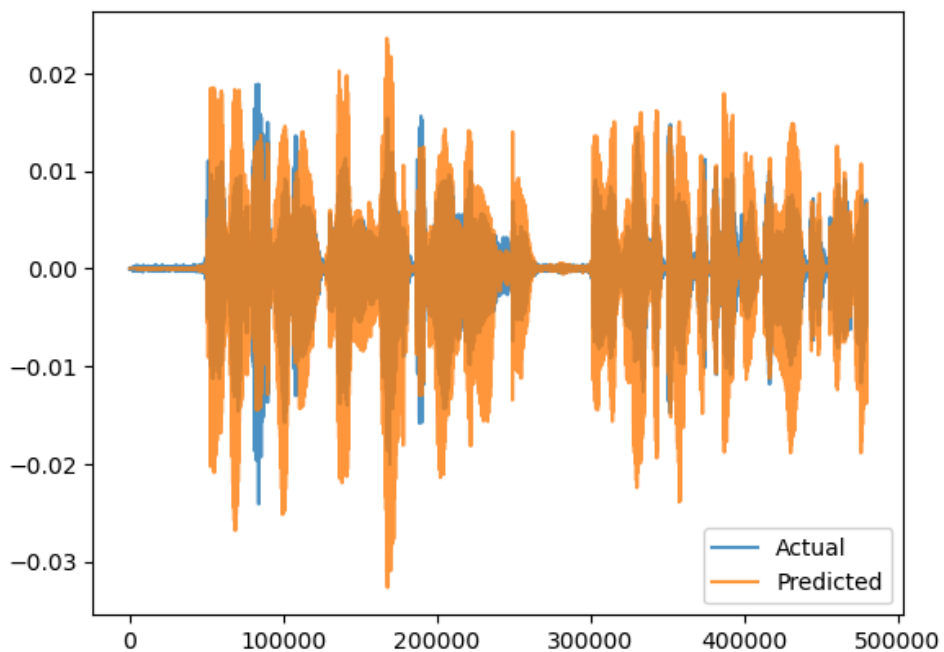


- Delayed optimal filter



Predicted vs actual panel contribution

- $e_{\text{cancel}} - e_{\text{no_cancel}}$ is what the panel added
- Compare to $ir * -g(w * x)$ where -g is cancel gain
- Had to align in time since $e_{\text{cancel}} - e_{\text{no_cancel}}$ were recorded separately



- What is the best we can match given any choice of shift and scale?
- 90 lag, .466 scale

